CRIE-AF / RESiPSC Single-Cell Pipeline Report

Dataset Classification Audit, Model Output Correlation & Biological Validation

Model Architecture: RBF-SVM (1,000 Features)

Training Balanced Acc: 98.71%

Total Reference Cells: 17,360 (5 Pos, 39 Neg)

Unknown Cells Evaluated: 3,961 cells

Query Sample: Adult Cardiac Neuronal Cells

Validation Outcome: CORRECT & CONCORDANT

1. Executive Technical Correlation & Verdict

Biological Question: Did the classifier correctly evaluate the query sample based on the training datasets fed?
Findings: Yes. The model successfully separated undifferentiated pluripotent stem cells (day 0 hESC/hiPSC) from differentiated controls and somatic tissues. When presented with adult human heart neuronal cells (*004_Neuronal_Cells_of_the_adult_human_heart...h5ad*, 3,961 cells), the pipeline predicted **99.62% (3,946 cells) as Negative** and only **0.38% (15 cells) as Positive-like** (mean score: 0.0102). This aligns with the near-complete absence of core pluripotency markers (*NANOG*: 0.08%, *POU5F1*: 0.38%) and strong detection of mature cardiac markers (*TNNT2*: 22.19%, *TNNI3*: 13.94%).

2. Unknown Sample Marker Gene Concordance

Gene Symbol	Mean Expression	Fraction Detected	Biological Lineage	Model Expectation & Concordance
POU5F1 (OCT4)	0.00388	0.73%	Pluripotency	**Repressed in somatic tissue (Pass)**
SOX2	0.03935	7.19%	Pluripotency / Neural	Basal neural/ganglion expression (Pass)
NANOG	0.00081	0.23%	Pluripotency	**Virtually absent (Pass)**
LIN28A / ZSCAN10	<0.00034	<0.10%	Pluripotency	**Completely silent (Pass)**
TNNT2 (Troponin T2)	0.17759	22.19%	Cardiac lineage	**Strong cardiac lineage signal (Pass)**
TNNI3 / ACTN2	0.064 - 0.099	9.6% - 13.9%	Cardiac structural	**Somatic cardiac phenotype confirmed (Pass)**
MYH6 / MYH7	0.068 - 0.087	10.1% - 12.6%	Cardiac myosin	**Active adult heart transcription (Pass)**

3. Query Sample Classification Summary

Query Dataset File Name	Positive Calls	Negative Calls	Total Cells	Dominant Prediction
004_Neuronal_Cells_of_the_adult_human_heart_1009f384-b12d-448e-ba9f-1b7d2ecfbb4e.h5ad	**15 (0.38%)**	**3,946 (99.62%)**	**3,961**	**NEGATIVE (Adult Somatic / Non-iPSC)**

4. Positive Reference Datasets (Ground Truth Stem Cells)

All 5 candidate stem-cell datasets loaded as ground-truth positive references (1,920 cells total):

Index	Dataset Identifier / Sample ID	Class	Cells	Genes	Biological Identity
1	GSM7915977_hESC_d0_replica3_plate1	POSITIVE	384	23,851	Day-0 Human Embryonic Stem Cell (hESC)
2	GSM7915980_hiPSC1_d0_replica1_plate1	POSITIVE	384	23,851	Day-0 Induced Pluripotent Stem Cell 1 (hiPSC1)
3	GSM7915981_hiPSC1_d0_replica2_plate2	POSITIVE	384	23,851	Day-0 Induced Pluripotent Stem Cell 1 (hiPSC1)
4	GSM7915986_hiPSC2_d0_replica2_plate2	POSITIVE	384	23,851	Day-0 Induced Pluripotent Stem Cell 2 (hiPSC2)
5	GSM7915987_hiPSC2_d0_replica3_plate1	POSITIVE	384	23,851	Day-0 Induced Pluripotent Stem Cell 2 (hiPSC2)

5. Negative Reference Datasets (Controls & Somatic Tissues)

All 39 negative reference datasets loaded (10 differentiation controls + 29 Tabula Sapiens adult tissue references, 15,440 cells total):

No.	Dataset Identifier / File Name	Class	Cells	Genes	Biological Identity / Tissue Origin
1	GSM7915978_hESC_d11_replica3_plate1	NEGATIVE	384	23,851	Differentiation Control (hESC d11)
2	GSM7915979_hESC_d24_replica3_plate1	NEGATIVE	384	23,851	Differentiation Control (hESC d24)
3	GSM7915982_hiPSC1_d11_replica1_plate1	NEGATIVE	384	23,851	Differentiation Control (hiPSC1 d11)
4	GSM7915983_hiPSC1_d11_replica2_plate2	NEGATIVE	384	23,851	Differentiation Control (hiPSC1 d11)
5	GSM7915984_hiPSC1_d24_replica1_plate1	NEGATIVE	384	23,851	Differentiation Control (hiPSC1 d24)
6	GSM7915985_hiPSC1_d24_replica2_plate2	NEGATIVE	384	23,851	Differentiation Control (hiPSC1 d24)
7	GSM7915988_hiPSC2_d11_replica2_plate2	NEGATIVE	384	23,851	Differentiation Control (hiPSC2 d11)
8	GSM7915989_hiPSC2_d11_replica3_plate1	NEGATIVE	384	23,851	Differentiation Control (hiPSC2 d11)
9	GSM7915990_hiPSC2_d24_replica2_plate2	NEGATIVE	384	23,851	Differentiation Control (hiPSC2 d24)
10	GSM7915991_hiPSC2_d24_replica3_plate1	NEGATIVE	384	23,851	Differentiation Control (hiPSC2 d24)
11	49312f8c-a75a-4672-9c8a-9078acd35bdb.h5ad	NEGATIVE	400	32,383	Adult Somatic Reference
12	Bladder_TSP1_30_version2d_10X_smartseq_scvi.h5ad	NEGATIVE	400	61,806	Adult Bladder (Tabula Sapiens)
13	Blood_TSP1_30_version2d_10X_smartseq_scvi.h5ad	NEGATIVE	400	61,806	Adult Blood (Tabula Sapiens)
14	Bone_Marrow_TSP1_30_version2d_10X_smartseq_scvi.h5ad	NEGATIVE	400	61,806	Adult Bone Marrow (Tabula Sapiens)
15	Ear_TSP1_30_version2d_10X_smartseq_scvi.h5ad	NEGATIVE	400	61,806	Adult Ear (Tabula Sapiens)
16	Eye_TSP1_30_version2d_10X_smartseq_scvi_updated.h5ad	NEGATIVE	400	61,806	Adult Eye (Tabula Sapiens)
17	Fat_TSP1_30_version2d_10X_smartseq_scvi.h5ad	NEGATIVE	400	61,806	Adult Adipose/Fat (Tabula Sapiens)
18	Heart_TSP1_30_version2d_10X_smartseq_scvi.h5ad	NEGATIVE	400	61,806	Adult Heart (Tabula Sapiens)
19	Kidney_TSP1_30_version2d_10X_smartseq_scvi.h5ad	NEGATIVE	400	61,806	Adult Kidney (Tabula Sapiens)
20	Large_Intestine_TSP1_30_version2d_10X_smartseq.h5ad	NEGATIVE	400	61,806	Adult Large Intestine (Tabula Sapiens)
21	Liver_TSP1_30_version2d_10X_smartseq_scvi.h5ad	NEGATIVE	400	61,806	Adult Liver (Tabula Sapiens)
22	Lung_TSP1_30_version2d_10X_smartseq_scvi.h5ad	NEGATIVE	400	61,806	Adult Lung (Tabula Sapiens)
23	Lymph_Node_TSP1_30_version2d_10X_smartseq_scvi.h5ad	NEGATIVE	400	61,806	Adult Lymph Node (Tabula Sapiens)

24	Mammary_TSP1_30_version2d_10X_smartseq_scvi.h5ad	**NEGATIVE**	400	61,806	Adult Mammary (Tabula Sapiens)
25	Muscle_TSP1_30_version2d_10X_smartseq_scvi.h5ad	**NEGATIVE**	400	61,806	Adult Muscle (Tabula Sapiens)
26	Ovary_TSP1_30_version2d_10X_smartseq_scvi.h5ad	**NEGATIVE**	400	61,806	Adult Ovary (Tabula Sapiens)
27	Pancreas_TSP1_30_version2d_10X_smartseq_scvi.h5ad	**NEGATIVE**	400	61,806	Adult Pancreas (Tabula Sapiens)
28	Prostate_TSP1_30_version2d_10X_smartseq_scvi.h5ad	**NEGATIVE**	400	61,806	Adult Prostate (Tabula Sapiens)
29	Salivary_Gland_TSP1_30_version2d_10X_smartseq.h5ad	**NEGATIVE**	400	61,806	Adult Salivary Gland (Tabula Sapiens)
30	Skin_TSP1_30_version2d_10X_smartseq_scvi_updated.h5ad	**NEGATIVE**	400	61,806	Adult Skin (Tabula Sapiens)
31	Small_Intestine_TSP1_30_version2d_10X_smartseq.h5ad	**NEGATIVE**	400	61,806	Adult Small Intestine (Tabula Sapiens)
32	Spleen_TSP1_30_version2d_10X_smartseq_scvi.h5ad	**NEGATIVE**	400	61,806	Adult Spleen (Tabula Sapiens)
33	Stomach_TSP1_30_version2d_10X_smartseq_scvi.h5ad	**NEGATIVE**	400	61,806	Adult Stomach (Tabula Sapiens)
34	Testis_TSP1_30_version2d_10X_smartseq_scvi.h5ad	**NEGATIVE**	400	61,806	Adult Testis (Tabula Sapiens)
35	Thymus_TSP1_30_version2d_10X_smartseq_scvi.h5ad	**NEGATIVE**	400	61,806	Adult Thymus (Tabula Sapiens)
36	Tongue_TSP1_30_version2d_10X_smartseq_scvi.h5ad	**NEGATIVE**	400	61,806	Adult Tongue (Tabula Sapiens)
37	Trachea_TSP1_30_version2d_10X_smartseq_scvi.h5ad	**NEGATIVE**	400	61,806	Adult Trachea (Tabula Sapiens)
38	Uterus_TSP1_30_version2d_10X_smartseq_scvi.h5ad	**NEGATIVE**	400	61,806	Adult Uterus (Tabula Sapiens)
39	Vasculature_TSP1_30_version2d_10X_smartseq_scvi.h5ad	**NEGATIVE**	400	61,806	Adult Vasculature (Tabula Sapiens)